This is a fantastic topic for students because it bridges the gap between **"writing for humans"** (content) and **"writing for machines"** (SEO and data science).

The following exercise is designed for a 90-minute lab session. It moves from conceptual understanding to hands-on coding and validation.

Exercise: "Making the Web Readable"

Objective: Students will learn how to implement JSON-LD schema markup to transform a standard webpage into a Google "Rich Result."

Phase 1: The Concept (15 Minutes)

Before touching code, students must understand that search engines don't "see" images or text the way we do. They need a translator.

- **The Problem:** A search engine sees "4.5" on a page. Is that a price? A rating? The version of a software?
- **The Solution: Schema.org**. A shared vocabulary used by Google, Bing, and Yahoo to define what data means.
- **The Result: Rich Results.** These are enhanced search listings (stars, price tags, recipe cook times) that increase click-through rates.

Phase 2: Decoding the Syntax (20 Minutes)

Explain that we use **JSON-LD** (JavaScript Object Notation for Linked Data) because it's the industry standard and keeps the code organized.

The Anatomy of a Schema Script:

JSON

```
<script type="application/ld+json">
{
  "@context": "https://schema.org/",
  "@type": "Product",
  "name": "SuperTech Headphones",
  "brand": {
    "@type": "Brand",
    "name": "AudioMax"
  },
  "aggregateRating": {
    "@type": "AggregateRating",
    "ratingValue": "4.8",
    "reviewCount": "120"
  }
}
```

</script>

Phase 3: Practical Workshop (45 Minutes)

Task: Create a "Recipe" Rich Result

Students will act as web developers for a cooking blog. They must take the raw data below and turn it into a structured JSON-LD block.

The Raw Data:

- **Name:** Grandma's Famous Chocolate Chip Cookies
- **Author:** Chef Maria
- **Prep Time:** 15 minutes
- **Cook Time:** 10 minutes
- **Rating:** 5 stars (based on 42 reviews)
- **Image URL:** https://example.com/cookie.jpg

Step-by-Step Instructions:

1. **Navigate to the Schema.org Recipe Documentation.** Look at the properties required.
2. **Open a Text Editor** (Notepad++, VS Code, or even a simple Google Doc).
3. **Draft the Script:** Use the "Product" example above as a template, but change the @type to Recipe.
4. **Use the "Nesting" Technique:** Ensure the aggregateRating is nested properly inside the Recipe object.

Phase 4: Validation (10 Minutes)

Code is useless if it has a missing comma or bracket. Students must validate their work using professional tools.

1. Copy your completed script.
2. Go to the Google Rich Results Test.
3. Select **"Code"** instead of "URL" and paste your script.
4. **The Goal:** Get a "Green Checkmark" indicating the page is eligible for rich results.

Discussion Questions for Wrap-up:

- **Ethical Consideration:** If you lie in your schema (e.g., giving yourself 5 stars when you have 2), what happens? (Answer: Google may penalize the site for "Spammy Structured Data").
- **User Experience:** Why do users click on Rich Results more often than plain text links?
- **Automation:** In a real job, would you write this by hand for 1,000 products? (Answer: No, you'd use plugins or dynamic database hooks).